

Civil Engineering

Calculus and Differential Methods

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working and use correct units.

Section A

1. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

2. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C$$

3. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$\frac{dy}{dx} + P(x)y = Q(x)$$

4. State the meaning of the formula below and identify its main symbols. **(4 marks)**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Section B

5. Find the gradient function of a road profile. **(10 marks)**
6. Determine area under a drainage channel curve. **(10 marks)**
7. Apply Newton-Raphson method to find a root. **(10 marks)**
8. Create one realistic civil engineering problem based on this topic and solve it fully. **(20 marks)**